Domain Password Policy & Project Planning Documents

AccessGuard Health – Healthcare IAM Operations Simulation

Lab record: enforcing a password policy on HVMC.LAN and completing the planning documents for the design document

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Project Phase	Domain security configuration and project planning
Board Tasks	Tasks 4, 5, 11, 14 (planning documents) and Task 30 (password policy)
Document Version	1.0
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Lab Date	November 2, 2026

1. Purpose

This document records the work completed on November 2, 2026. It continues the Week 1 and Week 2 documentation and covers two things: (a) setting and testing the **domain password policy** on the Samba4 domain controller, with screenshots, and (b) finishing the written **planning documents** that make up the first version of the design document.

Everything here is a fictional training environment: no real patient data or real credentials are used.

2. Starting Point

The lab already had a working domain controller and a joined client workstation (see the Week 1 and Week 2 documents). The domain controller, **hvmc-dc01** (192.168.56.10), was already running in VirtualBox, so no new machines were needed.

Component	Detail
Domain controller	hvmc-dc01, 192.168.56.10, Samba4 Active Directory, DNS, Kerberos
Client workstation	hvmc-client01, 192.168.56.11, joined to HVMC.LAN (powered off for this work)
Tool used	`samba-tool` over SSH from the Mac Terminal

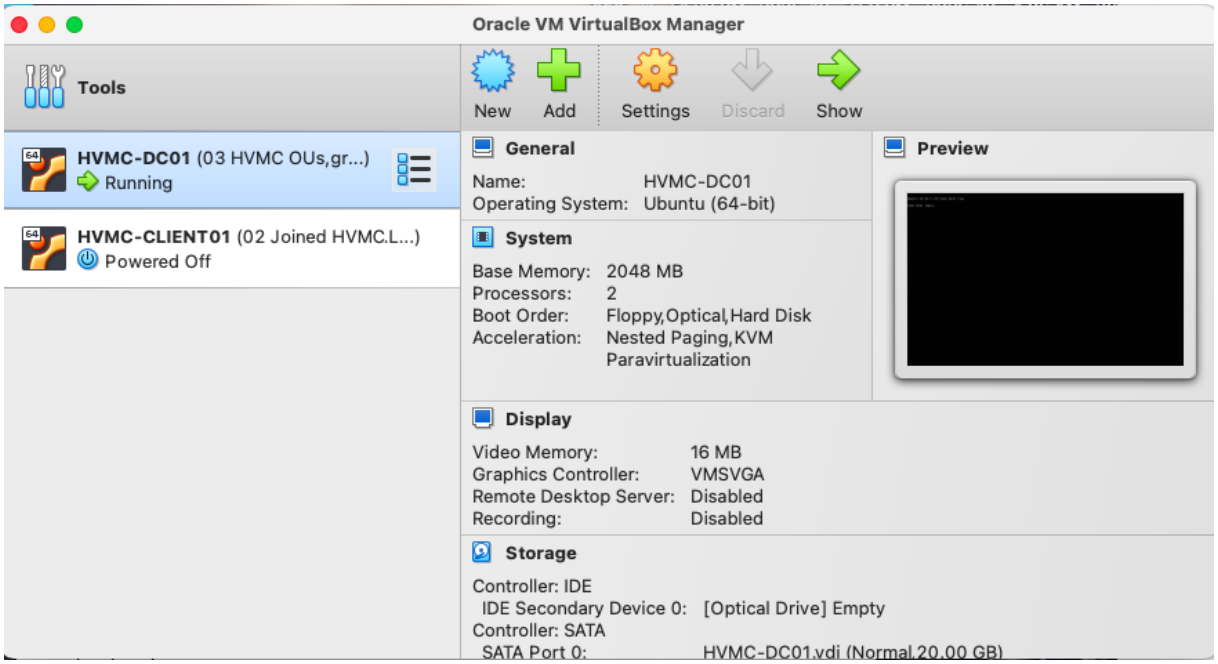


Figure 1: VirtualBox Manager: HVMC-DC01 running, HVMC-CLIENT01 powered off

3. Domain Password Policy (Board Task 30)

Goal: make accounts harder to guess and limit password-guessing attacks. In a hospital, a weak or guessable password could expose patient records. Samba4 uses `samba-tool` for domain password settings instead of a Windows Group Policy.

3.1 Check the current policy

After connecting by SSH, the current settings were displayed. This is the “before” record.

```
ssh ubuntu@192.168.56.10
sudo samba-tool domain passwordsettings show
```

The defaults left some gaps: the minimum length was only **7**, passwords lasted **42** days, and the lockout threshold was **0**, meaning an attacker could guess passwords without ever being locked out.

```
ubuntu@hvmc-dc01:~$ sudo samba-tool domain passwordsettings show
[sudo] password for ubuntu:
Password information for domain 'DC=hvmc,DC=lan'

Password complexity: on
Store plaintext passwords: off
Password history length: 24
Minimum password length: 7
Minimum password age (days): 1
Maximum password age (days): 42
Account lockout duration (mins): 30
Account lockout threshold (attempts): 0
Reset account lockout after (mins): 30
ubuntu@hvmc-dc01:~$
```

Figure 2: Before: default policy (length 7, maximum age 42 days, lockout threshold 0)

3.2 Change the policy

Three settings were changed, one command at a time, and each command confirmed success.

Setting	Before	After	Command
Minimum length	7	12	`--min-pwd-length=12`

Setting	Before	After	Command
Lockout threshold	0 (off)	5 attempts	`--account-lockout-threshold=5`
Maximum password age	42 days	90 days	`--max-pwd-age=90`
Complexity	on	on (unchanged)	–

```
sudo samba-tool domain passwordsettings set --min-pwd-length=12
sudo samba-tool domain passwordsettings set --account-lockout-threshold=5
sudo samba-tool domain passwordsettings set --max-pwd-age=90
```

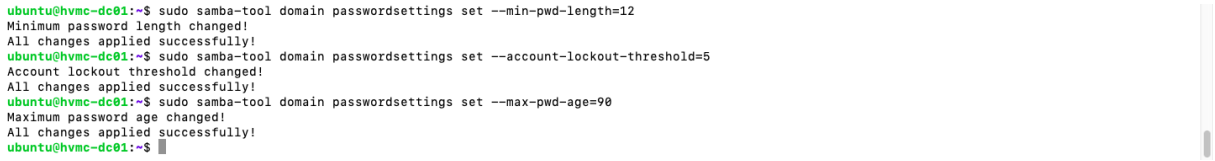


Figure 3: The three changes, each reporting "All changes applied successfully!"

3.3 Verify the new policy

The same show command was run again as the "after" record.

```
sudo samba-tool domain passwordsettings show
```

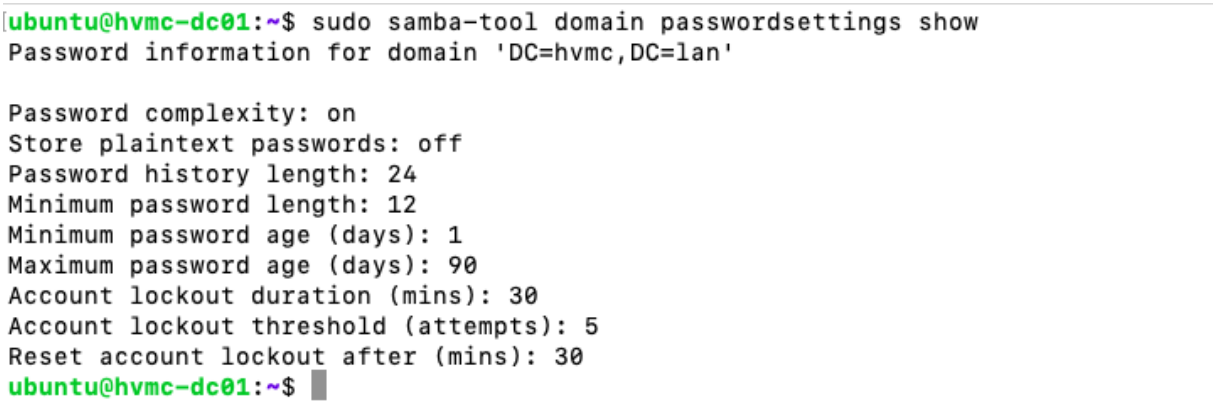


Figure 4: After: length 12, maximum age 90 days, lockout after 5 attempts, complexity on

3.4 Test that it is enforced

A policy is only useful if it is enforced. An account with the weak password `abc123` was requested, and the domain controller refused it. No account was created, so nothing had to be cleaned up.

```
sudo samba-tool user create pwtest1 abc123
```

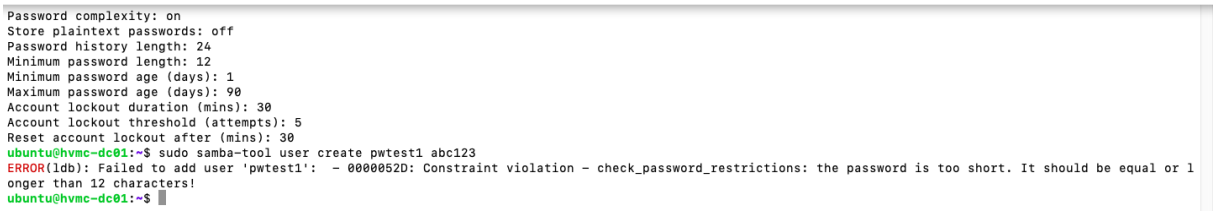


Figure 5: Weak password rejected: "the password is too short. It should be equal or longer than 12 characters!"

4. Planning Documents Completed

The written deliverables for the first version of the design document were finished. Each one is a Markdown file for the GitHub repository.

Board task	Document	What it contains
Task 4 – Research why IAM matters in healthcare	Why_IAM_Matters_in_Healthcare.md	Five-point problem statement and source list: IBM 2025 average healthcare breach cost of $7.42 million (highest of any industry for the 14th year, 279 days to contain); more than 35 million people affected by breaches reported to HHS in 2025; HIPAA Security Rule access control (45 CFR 164.312).
Task 5 – Draft project title and summary	Design_Document_v0.1.md	Title page, project summary and problem statement for “AccessGuard Health”.
Task 11 – Define MVP activities and scope	MVP_Activities_and_Scope.md	Five MVP activities (onboarding, transfer, termination, MFA support, quarterly access review), what is in scope and what is out of scope.
Task 14 – Finalize goals, objectives and key features	Goals_Objectives_and_Key_Features.md	Three goals, six objectives (three already completed) and the key features.

5. Project Board Progress

Measure	Before today	After today
Tasks completed	13	18
Tasks pending	53	48

Tasks marked **Completed** during this work:

- Task 4 – Research Why IAM Matters in Healthcare
- Task 5 – Draft Project Title and Summary
- Task 11 – Define MVP Activities and Scope
- Task 14 – Finalize Goals, Objectives and Key Features
- Task 30 – Configure Domain Password Policy

6. Troubleshooting Log

Issue	Cause	Resolution
Task wording said "via Group Policy"	Samba4 does not use Windows Group Policy for domain password settings	Used `samba-tool domain passwordsettings`, which the board deliverable also names
Pasted SSH command failed ("00~ssh: command not found")	Pasting added stray characters in Terminal	Typed the command by hand

7. Security Considerations

- All names, departments and users are **fictional**; no real patient or employee data is used.
- A **12-character minimum** and complexity make guessing and cracking harder; **lockout after 5 attempts** limits password-guessing attacks; a **90-day maximum age** limits how long a stolen password stays useful.
- The weak-password test used a throw-away name and created no account.
- Passwords entered during this work are practice-only credentials for an isolated lab; none appear in the screenshots.
- Policy values are lab choices that follow common practice and the HIPAA requirement to verify and limit access; production values should come from the organization's own policy.

8. Key Outcomes

- Closed a real security gap: account lockout was off and passwords could be as short as 7 characters.
- Proved the policy works with a rejected weak password.
- Completed the planning documents that form design document v0.1.
- Moved the project board from 13 to 18 completed tasks.

9. Remaining Work

Area	Pending tasks
Weeks 1–2 housekeeping	Task 6 (Mac setup check), Task 7 (Dell free space), Tasks 10 and 17 (status updates)
On-premises build	Tasks 38, 40, 43 (onboarding, transfer, termination), 46 (contractor expiration), 47 (automation scripts), 49 (access review), 26 (network diagram)
Cloud identity (Microsoft Entra ID)	Tasks 25, 33–36 (tenant, users, MFA, login), 44 (MFA support), 51 (Conditional Access), 56 (screenshots)
Documentation and portfolio	Tasks 39, 41, 45, 50 (tickets and report), 52–53 (SOPs and polish), 55, 57–58 (screenshots and portfolio), 60–67 (final review, publishing and closure)
Network Security 1	Firewall, IDS/IPS, ACL, Wireshark, scanner, penetration test, wireless and incident report (see the rubric alignment document)

10. Deliverables

- Domain Password Policy & Project Planning Documents: Lab Documentation for GitHub Portfolio (this document).
- Why_IAM_Matters_in_Healthcare.md, Design_Document_v0.1.md, MVP_Activities_and_Scope.md and Goals_Objectives_and_Key_Features.md.
- Updated project board with Status column (18 Completed, 48 Pending).